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Full Length Article

Data-driven-based fully distributed event-triggered control for nonlinear multi-agent systems ☆

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ABSTRACT

This paper investigates the fully distributed control issue for nonlinear multi-agent systems (MASs) under the limited network bandwidth. A novel event-triggered MFAC framework is developed to ensure the consensus of system output signals in a fully distributed manner, in which an adaptive step-size operator is introduced to eliminate the reliance on communication topology information. To save communication resources efficiently, a dynamic event-triggered mechanism (ETM) with switch-adjustable threshold parameters and dormancy waking functions is designed. This triggering mechanism not only accelerates the convergence of consensus errors in unstable situations, but also prolongs the inter-execution time while maintaining the desired control performance. In the entire control process, only measured input/output data are utilized. Through mathematical analysis, the average consensus errors of closed-loop MASs are proven to converge to zero asymptotically. Finally, the effectiveness and superiority of the proposed method are demonstrated through simulation comparisons.

1. Introduction

Over the past decade, multi-agent systems (MASs) composed of multiple agents have garnered significant attention from various fields, including autonomous vehicles [1–3], sensor networks [4–6], and power systems [7–9]. As a significant control issue in MASs, the consensus problem was first investigated in management science and statistics [10], and gradually became the foundation of distributed computing. Recently, a delay-classification-based distributed control method has been developed in [11] to address the leader-following consensus problem in MASs with large time-varying output delays. Zhu et al. proposed a distributed predefined-time control strategy in [12], in which neural network techniques were exploited to approximate the unknown dynamics of nonlinear systems. The authors in [13] addressed a fixed-time consensus control issue for the systems with full state constraints by using hyperbolic tangent and saturation functions.

On the other hand, the limited network bandwidth and computational resources have a negative impact on the communication efficiency of the systems [14,15]. The event-triggered control (ETC) strategy has emerged as an effective way to tackle this problem, ensuring that network resource is only utilized as “need” [16–18]. In [16], the event-triggered impulse control method is formulated

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to reduce communication frequency for stochastic systems. The event-based coordination control problem is investigated in [17] for MASs subject to external disturbances, in which the Zeno behaviors characterized by unlimited triggers within a finite time are excluded. In the subsequent work [18], the stability conditions for switched affine systems with static event-triggered sampling are derived, and the time intervals for controller updates are proven to be positive. To further extend inter-execution time, some dynamic/adaptive ETMs have been designed in [19–21] for single-network control systems or MASs. In [21], a distributed resilient event-triggering load frequency control method is proposed for power systems to defend against cyber attacks, in which a trigger condition with adjustable threshold parameters is exploited to reduce the communication frequency. However, the proposed triggering strategy primarily focuses on communication costs in the up-grading stage, which may lead to excessive communication burden in the down-grading case.

Note that the above works require accurate system modeling, which is difficult or even impractical to achieve in many real-world scenarios. To address this problem, some advanced fuzzy/neural network control methods have been proposed in [22–24] to approximate the unknown dynamics of systems. Despite these fuzzy/neural network algorithms that can effectively approximate the unknown dynamics of systems, they require more expert knowledge to achieve better control performance. Model-free adaptive control (MFAC) algorithms can efficiently circumvent these problems by establishing a dynamic linearization model based on input/output (I/O) data for nonlinear systems [25]. More recently, the MFAC method has been employed in [26,27] to address tracking issues in nonlinear multi-agent systems. To be specific, an iterative learning approach is applied to nonlinear networks in [26] to achieve consensus tracking control. In [27], a resilient MFAC framework with a predictive compensation mechanism is adopted for nonlinear MASs to mitigate the adverse effects of jamming attacks, ensuring that tracking errors converge to zero.

From the perspective of MASs, the distributed MFAC algorithms presented in [25–27] require the knowledge of Laplacian matrices, which limits their applications in large-scale systems, as it is difficult for each agent to obtain such global information. For this reason, a fully distributed MFAC algorithm is developed in [28] with the utilization of improved dynamic linearization techniques, in which an observer is employed to estimate the distributed output for updating the local controller. However, the produced observation errors enlarge the convergence range of system consensus errors, and the step-size operator of the controller needs to be re-adjusted to maintain satisfactory control performance when implementing the control algorithm in different physical systems. Besides, how to design an efficient event-triggered MFAC strategy to quickly restore control performance when the distributed systems become unstable has not been investigated in [28].

In this paper, we propose an MFAC method with a novel dynamic ETM to achieve consensus for nonlinear MASs in a fully distributed manner. To enhance the performance of ETM, a switchable threshold parameter is introduced in the ETC condition, which can be adjusted online to increase/decrease the data transmission frequency according to the system operation situation. Then, the event-triggered-based consensus protocol is utilized in the MFAC algorithms to achieve consensus among agents while saving communication resources for systems. To eliminate the reliance on global network information, an adaptive step-size operator that updates with local information is incorporated into the proposed MFAC framework. Moreover, an auxiliary factor is added to the controller to alleviate the negative impact of time-varying step-size on system stability. The major contributions of this paper are as follows:

1) A fully distributed event-triggered MFAC framework is constructed for the nonlinear MASs with an undirected and connected communication topology. In this framework, the control algorithms are driven by measurement I/O data and do not require knowledge of the systems' structure and parameters in practical applications.

2) A novel distributed control algorithm is first developed with adaptive step-size update laws to realize the consensus among agents in a fully distributed manner. Different from the distributed results in [25–27,29–32], the designed step-size factor is independent of the Laplacian matrix information and can be adaptively adjusted online.

3) To conserve communication resources more effectively, a dynamic ETM with switch-adjustable threshold parameters is established for distributed networks. This transmission mechanism not only promotes system convergence in undesirable situations, but also extends the overall inter-execution time when systems operate under normal conditions.

The rest of this paper is structured in the following manner: Preliminaries and problem formulation are detailed in Section 2. Section 3 presents the learning-based distributed MFAC algorithms that incorporate ETM. Section 4 offers the relevant stability analysis. Simulation tests are demonstrated in Section 5. Lastly, this article is concluded in Section 6.

2. Preliminaries and problem formulation

Let R denote the set of real numbers. The topology graph $\mathscr{G} = (\mathscr{V}, \mathscr{E})$ represents how information is exchanged within distributed networks, where $\mathscr{V} = \{v_1, v_2, \ldots, v_M\}$ is the set of vertices, and $\mathscr{E} \subseteq \mathscr{V} \times \mathscr{V}$ is the set of edges. An undirected graph $\mathscr{G}$ is said to be connected if there exists a path between every pair of distinct nodes. The adjacency matrix $\mathscr{A}$ is formed by the elements a_{ij}, where $a_{ii} = 0, a_{ij} = 1$ if $(v_i, v_j) \in \mathscr{E}$, and $a_{ij} = 0$ otherwise. The set $N_i = \{j \in \mathscr{V} \mid (v_j, v_i) \in \mathscr{E}\}$ represents the neighbors of node v_i.

For single-input single-output (SISO) nonlinear MASs, the dynamics of each agent are given as follows:

$$y_i(s+1) = f_i\left(y_i(s), u_i(s)\right), \quad i = 1, 2, \ldots, M \tag{1}$$

where s represents the discrete-time instant. $y_i(s) \in R$ denotes the output of agent i while $u_i(s) \in R$ represents its input. $f_i(\cdot)$ is an unknown nonlinear function.

The following assumptions about system dynamics and communication topology are given to facilitate controller design:

Assumption 1 *([26])*. The function $f_i(\cdot)$ has a continuous and positive partial derivative with respect to $u_i(s)$.

Assumption 2. The system (1) fulfills the Lipschitz conditions, namely,

$$\left|\Delta y_i(s+1)\right| \le b\left|\Delta u_i(s)\right|, \quad \Delta u_i(s) \neq 0,$$

where $\Delta y_i(s+1) = y_i(s+1) - y_i(s)$, $\Delta u_i(s) = u_i(s) - u_i(s-1)$, and b is a positive constant.

Assumption 3. The communication topology is modeled as an undirected and connected graph $\mathscr{G}$.

Lemma 1 *([25]). If the agent's dynamics (1) satisfy Assumptions 1-2, there is a variable $\psi_i(s)$, known as a pseudo-partial-derivative satisfying $\left|\psi_i(s)\right| \le b \quad (b>0)$, such that the nonlinear system (1) can be transformed into the following equivalent model:*

$$\Delta y_i(s+1) = \psi_i(s)\Delta u_i(s). \tag{2}$$

Remark 1. Assumptions 1-2 are acceptable for the nonlinear MASs. The conditions in Assumption 1 are common constraints in designing nonlinear controllers, while Assumption 2 indicates that finite variations in the input signal do not cause infinite changes in the output signal. Assumption 3 is a fundamental requirement for the communication topology, serving as a premise for applying the proposed control method.

The main control objective of this article is to develop an event-based fully distributed MFAC framework for the nonlinear MASs to realize consensus among agents without involving any global network information. Furthermore, a dynamic ETM will be designed to further minimize data transmission frequency while ensuring the desired control performance.

3. Event-based fully distributed MFAC law

Compared with the periodic transmission scheme, the ETM can reduce "unnecessary" data transfers by predefining trigger conditions that determine when to transmit sample data. To alleviate communication burden, many static ETMs with constant threshold parameters are utilized in [16–18,33–35]. However, simulation tests in these studies show that a large amount of data is still transmitted through the network even if the system is in stable operation, resulting in unnecessary waste of communication resources. To enhance communication efficiency, we propose the following dynamic ETM:

$$s_{l+1}^i \triangleq \inf\left\{s > s_l^i \mid h_i(s) \ge 0\right\}, \tag{3}$$

where s_l^i represents the lth triggering instant of the ith subsystem, and the corresponding time sequence $s_0^i, s_1^i, \cdots$ is decided by the event-triggered function $h_i(s)$ constructed as

$$h_i(s) = -\left(\sum_{j=1}^{M} a_{ij}\left(\check{y}_i(s) - \check{y}_j(s)\right)\right)^2 + \varpi_i(s_l^i)\rho_i(s)\sum_{j=1}^{M} a_{ij}\xi_i^2(s) - \mu e^{-\omega s}, \tag{4}$$

where $\check{y}_i(s) = y_i(s_l^i)\ \forall s \in [s_l^i, s_{l+1}^i)$ represents the system output at event-triggered instants, and $\xi_i(s) = \check{y}_i(s) - y_i(s)$ is exploited to describe the measurement error between $\check{y}_i(s)$ and $y_i(s)$. The tunable parameters $\mu > 0$ and $\omega > 0$ are employed to enhance the flexibility of the ETM. $\rho_i(s)$ represents the controller parameter to be designed later, while $\varpi_i\left(s_l^i\right)$ denotes the triggering threshold parameter.

Let $C[x_i(s)] = [x_1(s), x_2(s), \cdots, x_M(s)]^T$ and $D(x_i(s)) = diag(x_1(s), x_2(s), \cdots, x_M(s))$ denote the column vector and the diagonal matrix of the variable $x_i(s)$, respectively. The average consensus error of the ith agent is defined as $\delta_i(s) = y_i(s) - \frac{1}{M}\sum_{j=1}^{M} y_j(s)$, and it is easy to obtain

$$\delta(s) = H y(s), \tag{5}$$

where $\delta(s) = C[\delta_i(s)]$, $y(s) = C[y_i(s)]$. Let I_M denote the identity matrix of size M and 1_M represent a column vector with all entries equal to one. Then, the matrix H can be expressed as $H = I_M - \frac{1_M 1_M^T}{M}$. Define the consensus protocol as $\gamma_i(s) = \sum_{j=1}^{M} a_{ij}\left(y_i(s) - y_j(s)\right)$, and the signal received by the ith agent is given as

$$\check{\gamma}_i(s) = \gamma_i(s_l^i) = \sum_{j=1}^{M} a_{ij}\left(\check{y}_i(s) - \check{y}_j(s)\right). \tag{6}$$

With utilization of the signal $\gamma_i(s_l^i)$, the threshold parameter $\varpi_i\left(s_l^i\right) \in \left[\underline{\varpi}, \overline{\varpi}\right]$ is adjusted online in two cases:
1) when $\left|\gamma_i\left(s_l^i\right)\right| - \left|\gamma_i\left(s_{l-1}^i\right)\right| > 0$,

$$\varpi_i\left(s_l^i\right) = \varpi_i\left(s_{l-1}^i\right)\left(1 + \frac{2}{\pi}\arctan\left(\frac{\left|\gamma_i\left(s_l^i\right) - \gamma_i\left(s_{l-1}^i\right)\right|}{\left|\gamma_i\left(s_l^i\right)\right|}\right)\right), \tag{7}$$

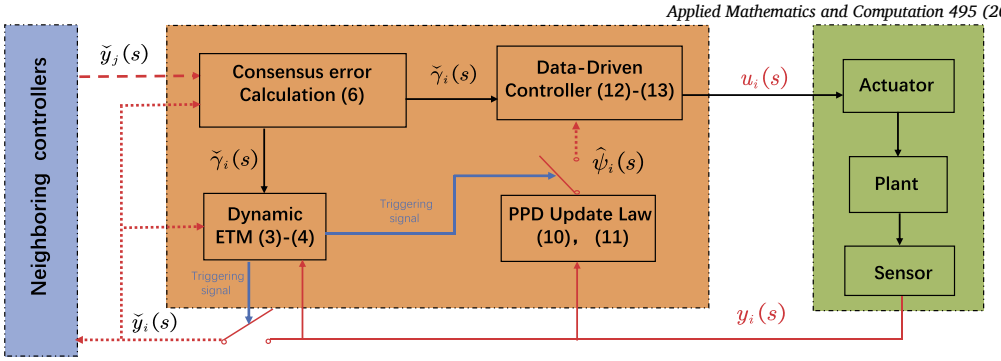


Fig. 1. The structure of the proposed dynamic event-triggered MFAC strategy.

2) when $\left|\gamma_i\left(s_l^i\right)\right| - \left|\gamma_i\left(s_{l-1}^i\right)\right| \leqslant 0$,

$$\begin{cases} \varpi_i\left(s_l^i\right) = \Theta\left(s_l^i\right), & \text{if } \Theta\left(s_l^i\right) \geqslant \underline{\varpi}, \\ \varpi_i\left(s_l^i\right) = \underline{\varpi}, & \text{otherwise}, \end{cases} \tag{8}$$

where $\Theta\left(s_l^i\right) = \varpi_i\left(s_{l-1}^i\right)\left(1 - \frac{2}{\pi}\arctan\left(\frac{\left|\gamma_i\left(s_l^i\right) - \gamma_i\left(s_{l-1}^i\right)\right|}{\left|\gamma_i\left(s_l^i\right)\right|}\right)\right)$ and $\underline{\varpi} > 0$ is an appropriate constant. In addition, an index operator $o(s)$ is introduced to denote the event occurrence, and it has

$$o(s) = \begin{cases} 1, & \text{event is triggered}, \\ 0, & \text{event is not triggered}. \end{cases} \tag{9}$$

Then, the fully distributed MFAC algorithms based on the dynamic ETM (3)-(4) for nonlinear systems (1) are designed as:

$$\hat{\psi}_i(s) = \hat{\psi}_i(s-1) + o(s)\frac{\eta \Delta u_i(s-1)}{\nu + \left|\Delta u_i(s-1)\right|^2} \times \left(\Delta y_i(s) - \hat{\psi}_i(s-1)\Delta u_i(s-1)\right), \tag{10}$$

$$\hat{\psi}_i(s) = \hat{\psi}_i(1), \text{ if } \left|\hat{\psi}_i(s)\right| \leqslant \sigma, \text{ if } \left|\Delta u_i(s-1)\right| \leqslant \sigma \text{ or } \operatorname{sign}\left(\hat{\psi}_i(s)\right) \neq \operatorname{sign}\left(\hat{\psi}_i(1)\right), \tag{11}$$

$$u_i(s) = u_i(s-1) + \frac{\rho_i(s)\hat{\psi}_i(s)}{\lambda + \left|\hat{\psi}_i(s)\right|^2}\frac{1}{\chi_i(s)}\sum_{j=1}^{M} a_{ij}\left(\breve{y}_j(s) - \breve{y}_i(s)\right), \tag{12}$$

$$\rho_i(s) = \rho_i(s-1) - \frac{\Delta y_i(s)\breve{\gamma}_i(s-1)}{2\rho_i(s-1)}, \tag{13}$$

where $\hat{\psi}_i(s)$ denotes the estimated value of $\psi_i(s)$ and it has $\hat{\psi}_i(s) = \hat{\psi}_i\left(s_l^i\right)$. $\eta \in (0,1]$ represents the step-sizes. ν and λ are positive weight factors. $\rho_i(s)$ is the time-varying step-size factor with $\rho_i(0)$ being a positive constant, while σ represents a sufficiently small positive number. $\chi_i(s)$ is the auxiliary factor satisfying $\chi_i(s) = 1 + \rho_i^2(s) + \breve{\gamma}_i^2(s)$.

The structure of the proposed control strategy is depicted in Fig. 1. When the event-triggered condition is violated, the data $\breve{y}_i(s)$ is transmitted to neighboring agents, and the PPD $\hat{\psi}_i(s)$ is sent to the controller. Subsequently, the control signal $u_i(s)$ is computed using both the local data and the $\breve{y}_j(s)$ received from neighboring agents. Throughout this process, only the measured output and the input signal are utilized. In other words, the control algorithms in (10)-(13) are data-driven.

Remark 2. It is worth noting that the step-size factors ρ_i in existing distributed MFAC algorithms [25–27,29–32] are the constants associated with the Laplacian matrices of network topologies and system characteristics. This implies that each agent must access to the global communication graph $\mathcal{G}$ to determine its local control parameters. However, manually selecting and adjusting step-size operators is challenging when applying distributed algorithms to large-scale MASs with diverse communication graphs. To better exhibit the advantages of the MFAC method in distributed networks, an adaptive step-size operator $\rho_i(s)$ is introduced into the distributed algorithms (10)-(13), which can be updated online without requiring any global network information and automatically converges to an appropriate value. Therefore, the devised control algorithms can be executed in a fully distributed manner.

Remark 3. As shown in (7)-(8), $\arctan(\cdot)$ function is employed to update the adaptive threshold parameter $\varpi_i\left(s_l^i\right)$ for the dynamic ETM (3)-(4), aiming to save the communication resource more effectively. When $\left|\gamma_i\left(s_l^i\right)\right| > \left|\gamma_i\left(s_{l-1}^i\right)\right|$, it indicates undesirable operation conditions and instability of the MASs, calling for more frequent data transmission to achieve consensus among the systems quickly. Therefore, it requires $\left|\varpi_i\left(s_l^i\right)\right| > \left|\varpi_i\left(s_{l-1}^i\right)\right|$ to speed up the trigger condition (3), which is executed by (7). On the contrary, a lower value of $\varpi_i\left(s_l^i\right)$ in (8) will lead to a lower communication frequency, thereby conserving more network resources. This strategy can not only rapidly mitigate the negative effects under unstable operation conditions by increasing the trigger frequency within

a short time, but also extend the internal execution time when systems are under conventional control. Furthermore, a supplemental condition $\varpi_i\left(s_l^i\right) = \underline{\varpi}$ in (8) is given to prevent a long time "dormancy" of the trigger scheme.

4. Stability analysis

In this section, it is proven that consensus among the subsystems of MASs (1) can be achieved by utilizing the proposed event-based fully distributed control algorithms.

Theorem 1. *Consider the nonlinear MASs (1) satisfying Assumptions 1-3. If the event-based fully distributed MFAC algorithms (10)-(13) are implemented, then there exists a $\lambda > \lambda_{\min} > 0$ such that the consensus error $\delta_i(s)$ of the systems converges to 0 as $s \to \infty$. Furthermore, the Zeno behavior of the dynamic ETM (3)-(4) is also excluded.*

Proof. First, the control protocol (12) is rearranged in a compact form as

$$u(s) = u(s-1) - \rho(s)F_1(s)\chi^{-1}(s)L_D\check{y}(s), \tag{14}$$

where $u(s) = C[u_i(s)]$, $\check{y}(s) = C[\check{y}_i(s)]$, $\rho(s) = D(\rho_i(s))$, $F_1(s) = D(\frac{\hat{\psi}_i(s)}{\lambda+|\hat{\psi}_i(s)|^2})$, and $\chi(s) = D(\chi_i(s))$. Recalling the converted linear model in (2) and the control algorithm (12), the average consensus error $\delta(s)$ in (5) can be rewritten as

$$\begin{aligned}\delta(s+1) &= \delta(s) + H\Delta y(s+1)\\ &= \delta(s) + H\psi(s)\Delta u(s)\\ &= \delta(s) - H\psi(s)\rho(s)F_1(s)\chi^{-1}(s)\check{\gamma}(s)\\ &= \delta(s) - H\rho(s)\varsigma(s)\chi^{-1}(s)L_D\check{y}(s),\end{aligned} \tag{15}$$

where $\psi(s) = D(\psi_i(s)), \Delta u(s) = u(s) - u(s-1), \check{\gamma}(s) = C[\check{\gamma}_i(s)], \varsigma(s) = D(\varsigma_i(s)) = \psi(s)F_1(s)$ with the variable

$$\varsigma_i(s) = \frac{\psi_i(s)\hat{\psi}_i(s)}{\lambda + \left|\hat{\psi}_i(s)\right|^2}, \quad i = 1, 2, \dots M.$$

The rest of the proof is divided into three parts as follows: 1) the boundedness of the parameter estimation error $\tilde{\psi}_i(s) = \hat{\psi}_i(s) - \psi_i(s)$, 2) the convergence of the average consensus error $\delta_i(s)$, 3) the exclusion of Zeno behaviors.

Part (I) [The Boundness of the Estimation Error $\tilde{\psi}_i(s)$]:

According to the parameter update law (10) and Lemma 1, it is easy to obtain

$$\tilde{\psi}_i(s) = \left(1 - o(s)\frac{\eta\Delta u_i^2(s-1)}{\mu + \left|\Delta u_i(s-1)\right|^2}\right) \times \tilde{\psi}_i(s-1) + \psi_i(s-1) - \psi_i(s). \tag{16}$$

Taking the absolute value on (16), one has

$$\left|\tilde{\psi}_i(s)\right| \le \left|1 - o(s)\frac{\eta\Delta u_i^2(s-1)}{\mu + \left|\Delta u_i(s-1)\right|^2}\right| \left|\tilde{\psi}_i(s-1)\right| + \left|\psi_i(s-1) - \psi_i(s)\right|. \tag{17}$$

Due to $\left|\Delta u_i(s)\right| \neq 0$, by choosing suitable parameters $0 < \eta \le 1$ and $\mu \ge 0$, there is $c \in (0,1)$ such that

$$0 < \left|1 - o(s)\frac{\eta\Delta u_i^2(s-1)}{\mu + \left|\Delta u_i(s-1)\right|^2}\right| \le c < 1 \tag{18}$$

holds. Given that b is the upper bound of $\left|\psi_i(s)\right|$, it is not difficult to get $\left|\psi_i(s) - \psi_i(s-1)\right| \le 2b$. Then, we have

$$\left|\tilde{\psi}_i(s)\right| \le c\left|\tilde{\psi}_i(s-1)\right| + 2b \le \cdots \le c^{s-1}\left|\tilde{\psi}_i(1)\right| + \frac{2b\left(1 - c^{s-1}\right)}{1-c}, \tag{19}$$

which reveals that $\tilde{\psi}_i(s)$ is bounded. Obviously, $\hat{\psi}_i(s)$ is bounded for all $i = 1, 2, \dots, M$, as $|\psi_i(s)| \le b$, and there exists $\lambda > \lambda_{\min} > 0$ such that time-varying $\varsigma_i(s)$ satisfies

$$0 < \varsigma_i(s) = \frac{\psi_i(s)\hat{\psi}_i(s)}{\lambda + \left|\hat{\psi}_i(s)\right|^2} \le \frac{b\hat{\psi}_i(s)}{2\sqrt{\lambda}\left|\hat{\psi}_i(s)\right|} < \frac{b}{2\sqrt{\lambda_{\min}}} < 1. \tag{20}$$

Part (II) [The Convergence of Average Consensus Error $\delta_i(s)$]:

On the basis of control law (12) and parameter update law (13) for the system (2), we can acquire

$$\Delta\rho_i(s+1) = -\frac{\Delta y_i(s+1)\check{\gamma}_i(s)}{2\rho_i(s)} = \frac{\rho_i(s)\varsigma_i(s)\chi_i^{-1}(s)\check{\gamma}_i(s)\check{\gamma}_i(s)}{2\rho_i(s)} = \frac{\varsigma_i(s)\check{\gamma}_i^2(s)}{2\chi_i(s)} \geq 0. \tag{21}$$

It is obvious that the step-size factor $\rho_i(s) = \rho_i(0) + \sum_{k=1}^{s}\Delta\rho_i(k)$ is a positive parameter. Choose the quadratic Lyapunov function as

$$V(s) = V_1(s) + V_2(s),$$

where $V_1(s) = \delta^T(s)L_D\delta(s)$, $V_2(s) = \sum_{i=1}^{M}(\rho_i(s) - \alpha)^2$, and $\alpha > 0$ denotes the constant to be determined later.

Substitute (15) into the difference of the function $V_1(s)$ to yield

$$\begin{aligned}
&\Delta V_1(s+1)\\
=&V_1(s+1) - V_1(s)\\
=&\delta^T(s+1)L_D\delta(s+1) - \delta^T(s)L_D\delta(s)\\
=&\left[\delta(s) - H\rho(s)\varsigma(s)\chi^{-1}(s)L_D\check{y}(s)\right]^T L_D[\delta(s) - H\rho(s)\varsigma(s)\chi^{-1}(s)L_D\check{y}(s)] - \delta^T(s)L_D\delta(s)\\
\overset{(a)}{=}&-2\delta^T(s)L_D\left[\rho(s)\varsigma(s)\chi^{-1}(s)L_D\check{y}(s)\right] + \left[\check{y}^T(s)L_D^T\chi^{-1}(s)\varsigma(s)\rho(s)\right]L_D\left[\rho(s)\varsigma(s)\chi^{-1}(s)L_D\check{y}(s)\right]\\
\leqslant&-2\delta^T(s)L_D^T\left[\rho(s)\varsigma(s)\chi^{-1}(s)L_D\check{y}(s)\right] + \lambda_M\left[\check{y}^T(s)L_D^T\chi^{-1}(s)\varsigma(s)\rho(s)\right]\left[\rho(s)\varsigma(s)\chi^{-1}(s)L_D\check{y}(s)\right]\\
=&-2\sum_{i=1}^{M}\frac{\rho_i(s)\varsigma_i(s)}{\chi_i(s)}\sum_{j=1}^{M}a_{ij}(\delta_i(s) - \delta_j(s))\sum_{j=1}^{M}a_{ij}\left(\check{y}_i(s) - \check{y}_j(s)\right) + \lambda_M\sum_{i=1}^{M}\rho_i^2(s)\varsigma_i^2(s)\frac{\check{\gamma}_i^2(s)}{\chi_i^2(s)}\\
\overset{(b)}{\leq}&-2\sum_{i=1}^{M}\rho_i(s)\varsigma_i(s)\frac{\gamma_i(s)\check{\gamma}_i(s)}{\chi_i(s)} + \lambda_M\sum_{i=1}^{M}\varsigma_i(s)\frac{\check{\gamma}_i^2(s)}{\chi_i(s)},
\end{aligned} \tag{22}$$

where $\gamma(s) = C[\gamma_i(s)]$, and $\lambda_M > 0$ is the maximum eigenvalue of the matrix L_D. (a) follows from $H = I_M - \frac{1_M 1_M^T}{M}$ satisfying $L_D H = HL_D = L_D$. The inequality (b) follows from $\chi_i(s) = 1 + \rho_i^2(s) + \check{\gamma}_i^2(s)$ and (20), that is, $0 < \varsigma_i(s) < 1$, $0 < \frac{\rho_i^2(s)}{\chi_i(s)} = \frac{\rho_i^2(s)}{1+\rho_i^2(s)+\check{\gamma}_i^2(s)} < 1$. Since $a_{ij} = a_{ji}$ holds under the Assumption 3, we have

$$\sum_{i=1}^{M}\xi_i(s)\sum_{j=1}^{M}a_{ij}\left(\xi_i(s) - \xi_j(s)\right) = \frac{1}{2}\sum_{i=1}^{M}\sum_{j=1}^{M}a_{ij}\left(\xi_i(s) - \xi_j(s)\right)^2, \tag{23}$$

and

$$\sum_{i=1}^{M}\sum_{j=1}^{M}a_{ij}(\xi_i(s) - \xi_j(s))^2 \leq 2\sum_{i=1}^{M}\sum_{j=1}^{M}a_{ij}\xi_i^2(s) + 2\sum_{i=1}^{M}\sum_{j=1}^{M}a_{ij}\xi_j^2(s) = 4\sum_{i=1}^{M}\sum_{j=1}^{M}a_{ij}\xi_i^2(s). \tag{24}$$

By using the definitions of $\gamma_i(s)$, $\xi_i(s)$, and $\delta_i(s)$, it is easy to verify through Young's inequality that

$$\begin{aligned}
-\sum_{i=1}^{M}\gamma_i(s)\check{\gamma}_i(s) =& -\sum_{i=1}^{M}\left[\sum_{j=1}^{M}a_{ij}\left(\delta_i(s) - \delta_j(s)\right)\sum_{j=1}^{M}a_{ij}\left(\check{y}_i(s) - \check{y}_j(s)\right)\right]\\
\overset{(a)}{=}&\sum_{i=1}^{M}\left[-\left(\sum_{j=1}^{M}a_{ij}\left(\check{y}_i(s) - \check{y}_j(s)\right)\right)^2 + \sum_{j=1}^{M}a_{ij}\left(\xi_i(s) - \xi_j(s)\right)\sum_{j=1}^{M}a_{ij}\left(\check{y}_i(s) - \check{y}_j(s)\right)\right]\\
\leqslant&\sum_{i=1}^{M}\left[-\frac{1}{2}\left(\sum_{j=1}^{M}a_{ij}\left(\check{y}_i(s) - \check{y}_j(s)\right)\right)^2 + \frac{1}{2}\left(\sum_{j=1}^{M}a_{ij}\left(\xi_i(s) - \xi_j(s)\right)\right)^2\right]\\
\leqslant&-\frac{1}{2}\sum_{i=1}^{M}\check{\gamma}_i^2(s) + \frac{\beta}{2}\sum_{i=1}^{M}\xi_i(s)\sum_{j=1}^{M}a_{ij}\left(\xi_i(s) - \xi_j(s)\right)\\
\overset{(23)}{=}&-\frac{1}{2}\sum_{i=1}^{M}\check{\gamma}_i^2(s) + \frac{\beta}{4}\sum_{i=1}^{M}\sum_{j=1}^{M}a_{ij}\left(\xi_i(s) - \xi_j(s)\right)^2\\
\overset{(24)}{\leqslant}&-\frac{1}{2}\sum_{i=1}^{M}\check{\gamma}_i^2(s) + 2\beta\sum_{i=1}^{M}\sum_{j=1}^{M}a_{ij}\xi_i^2(s) - \frac{\beta}{4}\sum_{i=1}^{M}\sum_{j=1}^{M}a_{ij}\left(\xi_i(s) - \xi_j(s)\right)^2,
\end{aligned} \tag{25}$$

where β is a positive constant satisfying $\beta = \frac{||L_D||^2}{\lambda_2}$ with λ_2 being the minimum non-zero eigenvalue of L_D. The equality (a) follows from the fact that $\delta_i(s) - \delta_j(s) = \left(\check{y}_i(s) - \check{y}_j(s)\right) - \left(\xi_i(s) - \xi_j(s)\right)$.

Employing (25), the derivative of $V_1(s+1)$ becomes

$$\begin{aligned}
\Delta V_1(s+1) \leqslant & 2\sum_{i=1}^{M} \rho_i(s)\varsigma_i(s)\frac{\gamma_i(s)\check{\gamma}_i(s)}{\chi_i(s)} + \lambda_M \sum_{i=1}^{M} \varsigma_i(s)\frac{\check{\gamma}_i^2(s)}{\chi_i^2(s)} \\
\leqslant & -\sum_{i=1}^{M} \rho_i(s)\varsigma_i(s)\frac{\check{\gamma}_i^2(s)}{\chi_i(s)} + 4\beta \sum_{i=1}^{M} \frac{\rho_i(s)\varsigma_i(s)}{\chi_i(s)} \sum_{j=1}^{M} a_{ij}\xi_i^2(s) + \lambda_M \sum_{i=1}^{M} \varsigma_i(s)\frac{\check{\gamma}_i^2(s)}{\chi_i^2(s)} \\
& - \frac{\beta}{2} \sum_{i=1}^{M} \frac{\rho_i(s)\varsigma_i(s)}{\chi_i(s)} \sum_{j=1}^{M} a_{ij} \left(\xi_i(s) - \xi_j(s)\right)^2 \\
= & \sum_{i=1}^{M} \left(-\rho_i(s)\varsigma_i(s) + \lambda_M \varsigma_i(s)\right) \frac{\check{\gamma}_i^2(s)}{\chi_i(s)} + 4\beta \sum_{i=1}^{M} \frac{\rho_i(s)\varsigma_i(s)}{\chi_i(s)} \sum_{j=1}^{M} a_{ij}\xi_i^2(s) \\
& - \frac{\beta}{2} \sum_{i=1}^{M} \frac{\rho_i(s)\varsigma_i(s)}{\chi_i(s)} \sum_{j=1}^{M} a_{ij} \left(\xi_i(s) - \xi_j(s)\right)^2 .
\end{aligned} \tag{26}$$

From (13) and (21), we have

$$\begin{aligned}
\Delta V_2(s+1) = & \sum_{i=1}^{M} \left(\rho_i(s+1) - \alpha\right)^2 - \sum_{i=1}^{M} \left(\rho_i(s) - \alpha\right)^2 \\
= & \sum_{i=1}^{M} \left(\rho_i(s+1) + \rho_i(s) - 2\alpha\right) \left(\rho_i(s+1) - \rho_i(s)\right) \\
= & \sum_{i=1}^{M} \left(2\rho_i(s) - 2\alpha + \rho_i(s+1) - \rho_i(s)\right) \left(\rho_i(s+1) - \rho_i(s)\right) \\
= & \sum_{i=1}^{M} \left(2\rho_i(s) - 2\alpha + \frac{\varsigma_i(s)\check{\gamma}_i^2(s)}{2\chi_i(s)}\right) \frac{\varsigma_i(s)\check{\gamma}_i^2(s)}{2\chi_i(s)} \\
\leqslant & \sum_{i=1}^{M} \left(\rho_i(s)\varsigma_i(s) - \alpha\varsigma_i(s) + \frac{1}{4}\varsigma_i(s)\right) \frac{\check{\gamma}_i^2(s)}{\chi_i(s)}
\end{aligned} \tag{27}$$

where $0 < \varsigma_i(s) < 1$, $0 < \frac{\check{\gamma}_i^2(s)}{\chi_i(s)} = \frac{\check{\gamma}_i^2(s)}{1+\rho_i^2(s)+\check{\gamma}_i^2(s)} < 1$.

Based on the above equation and (26), it has

$$\begin{aligned}
\Delta V(s+1) = & \Delta V_1(s+1) + \Delta V_2(s+1) \\
= & \left(-\alpha + \lambda_M + \frac{1}{4}\right) \sum_{i=1}^{M} \varsigma_i(s)\frac{\check{\gamma}_i^2(s)}{\chi_i(s)} + 4\beta \sum_{i=1}^{M} \frac{\rho_i(s)\varsigma_i(s)}{\chi_i(s)} \sum_{j=1}^{M} a_{ij}\xi_i^2(s) \\
& - \frac{\beta}{2} \sum_{i=1}^{M} \frac{\rho_i(s)\varsigma_i(s)}{\chi_i(s)} \sum_{j=1}^{M} a_{ij} \left(\xi_i(s) - \xi_j(s)\right)^2 \\
= & -2C_1 \sum_{i=1}^{M} \varsigma_i(s)\frac{\check{\gamma}_i^2(s)}{\chi_i(s)} + 4\beta \sum_{i=1}^{M} \frac{\rho_i(s)\varsigma_i(s)}{\chi_i(s)} \sum_{j=1}^{M} a_{ij}\xi_i^2(s) \\
& - \frac{\beta}{2} \sum_{i=1}^{M} \frac{\rho_i(s)\varsigma_i(s)}{\chi_i(s)} \sum_{j=1}^{M} a_{ij} \left(\xi_i(s) - \xi_j(s)\right)^2
\end{aligned} \tag{28}$$

where $C_1 = (\frac{\alpha}{2} - \frac{\lambda_M}{2} - \frac{1}{8})$.

Select the parameter $\alpha > \frac{1}{4} + \lambda_M + \frac{8\beta}{\varpi}$ such that $C_1 > 0$ and $\frac{4\beta}{C_1 \varpi_i(s)} < 1$. Moreover, from event-triggered function (4) and the equation (6), one has

$$\lim_{s\to\infty} \Delta V(s+1)$$

$$
\begin{aligned}
&= \lim_{s\to\infty} -2C_1 \sum_{i=1}^{M} \varsigma_i(s)\frac{\check{\gamma}_i^2(s)}{\chi_i(s)} + 4\beta \sum_{i=1}^{M} \frac{\rho_i(s)\varsigma_i(s)}{\chi_i(s)} \sum_{j=1}^{M} a_{ij}\xi_i^2(s) - \frac{\beta}{2}\sum_{i=1}^{M}\frac{\rho_i(s)\varsigma_i(s)}{\chi_i(s)}\sum_{j=1}^{M} a_{ij}\left(\xi_i(s)-\xi_j(s)\right)^2 \\
&= \lim_{s\to\infty} \sum_{i=1}^{M} \frac{C_1\varsigma_i(s)}{\chi_i(s)}\left[-2\check{\gamma}_i^2(s) + \frac{4\beta}{C_1\varpi_i(s)}\varpi_i(s)\rho_i(s)\sum_{j=1}^{M} a_{ij}\xi_i^2(s) - \frac{\beta}{2C_1}\rho_i(s)\sum_{j=1}^{M} a_{ij}\left(\xi_i(s)-\xi_j(s)\right)^2\right] \\
&\leqslant \lim_{s\to\infty} \sum_{i=1}^{M} \frac{C_1\varsigma_i(s)}{\chi_i(s)}\left[-\check{\gamma}_i^2(s) - \left(\sum_{j=1}^{M} a_{ij}\left(\check{y}_i(s)-\check{y}_j(s)\right)\right)^2 + \varpi_i(s)\rho_i(s)\sum_{j=1}^{M} a_{ij}\xi_i^2(s) - \Theta(s)\right] \\
&\leqslant \lim_{s\to\infty} \sum_{i=1}^{M} \frac{C_1\varsigma_i(s)}{\chi_i(s)}\left(-\check{\gamma}_i^2(s) + \mu e^{-\omega s} - \Theta(s)\right) \\
&= \lim_{s\to\infty} \sum_{i=1}^{M} -\frac{C_1\varsigma_i(s)}{\chi_i(s)}\left(\check{\gamma}_i^2(s) + \Theta(s)\right) \\
&\leqslant 0
\end{aligned} \tag{29}
$$

where $\Theta(s) = \frac{\beta}{2C_1}\rho_i(s)\sum_{j=1}^{M} a_{ij}\left(\xi_i(s)-\xi_j(s)\right)^2 \geq 0$.

Based on (29), for $s \in [0,\infty)$, it can be found that

$$
-V(0) + \lim_{s\to\infty} C_1 \sum_{\iota=1}^{s}\sum_{i=1}^{M}\frac{\varsigma_i(\iota)}{\chi_i(\iota)}\left[\check{\gamma}_i^2(\iota)+\Theta(\iota)\right] \leq -V(0) - \lim_{s\to\infty}\sum_{\iota=1}^{s}\Delta V(\iota) = \lim_{s\to\infty} -V(s) \leq 0
$$

and

$$
\lim_{s\to\infty}\sum_{\iota=0}^{s}\sum_{i=1}^{M}\frac{\varsigma_i(\iota)}{\chi_i(\iota)}\left[\check{\gamma}_i^2(\iota)+\Theta(\iota)\right] \leq C_1^{-1}V(0).
$$

Due to $0<\varsigma_i(s)<1, 0<\chi_i(\iota)<1, 0<C_1$ and $0<\rho_i(\iota)$, it has

$$
\lim_{s\to\infty}\check{\gamma}_i^2(s) = \lim_{s\to\infty}\left[\sum_{j=1}^{M} a_{ij}\left(\check{y}_i(s)-\check{y}_j(s)\right)\right]^2 = 0,
$$

and

$$
\lim_{s\to\infty}\Theta(s) = \lim_{s\to\infty}\frac{\beta}{2C_1}\rho_i(s)\sum_{j=1}^{M} a_{ij}\left(\xi_i(s)-\xi_j(s)\right)^2 = 0,
$$

which implies $\lim_{s\to\infty}(\check{y}_i(s)-\check{y}_j(s)) = 0$, $\lim_{s\to\infty}\left(\xi_i(s)-\xi_j(s)\right)^2 = \lim_{s\to\infty}\left(\check{y}_i(s)-y_i(s)+y_j(s)-\check{y}_j(s)\right)^2 = 0$, then we can obtain $\lim_{s\to\infty} y_i(s) \to \lim_{s\to\infty} y_j(s)$ and $\lim_{s\to\infty}\delta_i(s)\to 0$.

Part(III) [The Exclusion of Zeno Behaviors]:

Next, we will prove that the network described in (1) does not demonstrate Zeno behavior under the event-based transmission scheme (3). Assume the event triggering instant decided by the function in (3) is $s_{l'}^i, l' \in R$, and Zeno behavior has been exhibited in the nonlinear system (1), which indicates $\lim_{l'\to\infty} s_{l'+1}^i = s_\infty^i < \infty$. At the instant $s_{l'+1}^i$, there is $\xi_i\left(s_{l'+1}^i\right) = 0$, and $h_i\left(s_{l'+1}^i\right) \geq 0$.

Following from (4), the function satisfies the inequality:

$$
\varpi_i\left(s_{l'+1}^i\right)\rho_i\left(s_{l'+1}^i\right)\sum_{j=1}^{M} a_{ij}\xi_i^2\left(s_{l'+1}^i\right) \geqslant \mu e^{-\omega s_{l'+1}^i},
$$

from which we can get

$$
0 = \varpi_i\left(s_{l'+1}^i\right)\rho_i\left(s_{l'+1}^i\right)\sum_{j=1}^{M} a_{ij}\xi_i^2\left(s_{l'+1}^i\right) \geqslant \mu e^{-\omega s_\infty^j}
$$

and it indicates that $s_\infty^i \to \infty$, which undoubtedly contradicts the previously stated conditions. As a result, no Zeno behavior will be demonstrated in the distributed systems under the proposed event-triggered conditions (3)-(4). In other words, it is Zeno-free for the nonlinear systems under ETM (3). This completes the proof. □

Remark 4. For the discrete-time system, the interval time between two adjacent sampling times is usually set very small to meet the compatibility of Shannon's theorem. Information transmission occurring at each sampling instant increases the communication

Table 1
Comparisons on event-triggered frequency.

Frequency / Topology / Control	I (Agents 1-3)	II (Agents 1-4)	III (Agents 1-4)	IV (Agents 1-4)
The proposed dynamic ETM	23% 26% 24%	26.5% 21.5% 22.5% 23%	47.5% 39% 20.5% 21.5%	38.5% 38% 22% 21.5%
The static ETM in [33–35]	31% 41.5% 39%	34% 26.5% 35% 33%	51.5% 42% 31% 30.5%	44% 43.5% 30.5% 29.5%

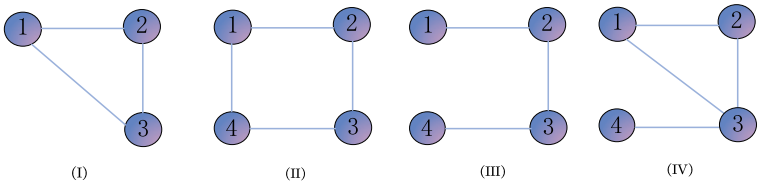
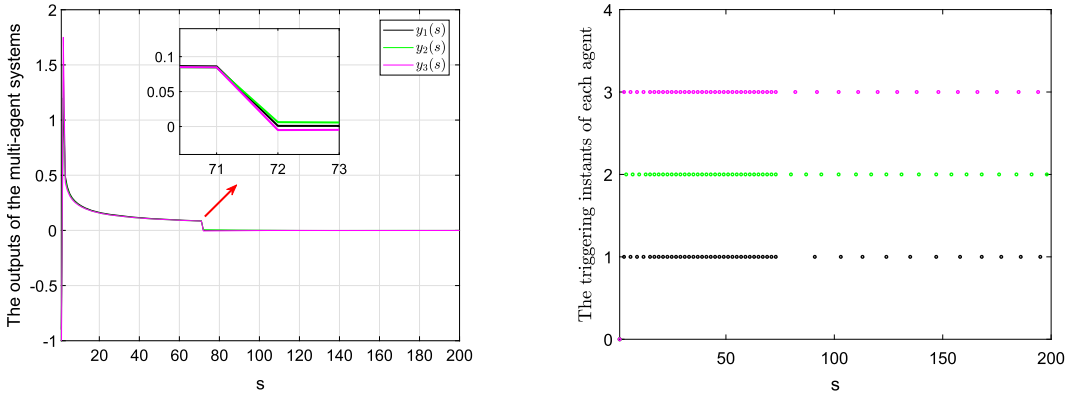


Fig. 2. The communication topology structures of the multi-agent systems.



(a) The output responses (b) The event-triggered instants

Fig. 3. The output responses of the MASs with the communication topology I.

burden of the systems. The exclusion of Zeno behaviors aims to avoid excessive communication frequency, which is not restricted to the control objective in this paper.

5. Simulation examples

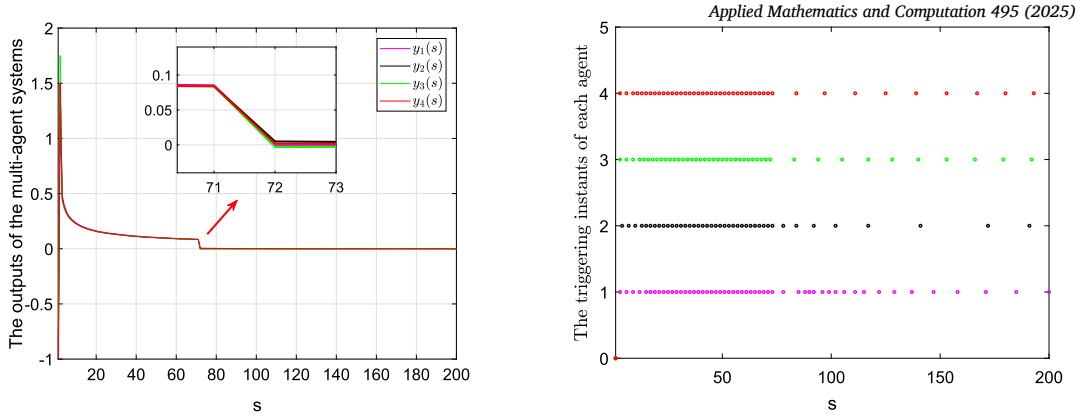
In this section, the effectiveness of the proposed approach is demonstrated through some numerical simulations conducted on the nonlinear systems consisting of three/four agents. The dynamics of each agent are given by (see in [26]):

$$y_i(s+1)=\begin{cases}\dfrac{y_i(s)}{1+y_i^2(s)}+u_i^2(s), & 0\le s\le 70\\ \dfrac{y_i(s)y_i(s-1)y_i(s-2)u_i(s-1)+g(s)u_i(s)}{1+y_i(s-1)^2+y_i(s-2)^2}, & 70<s\le 100\end{cases}$$

with a time-varying parameter $g(s)=1+\mathrm{rand}(s/50)$.

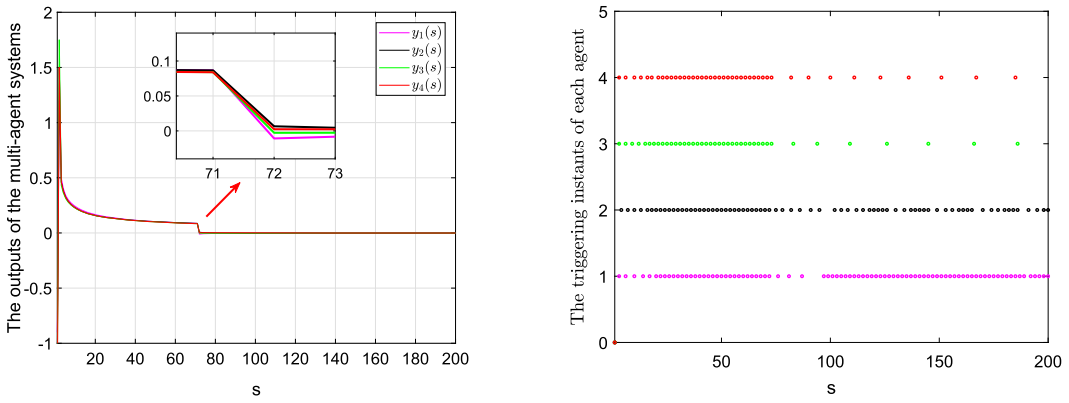
The initial values for the nonlinear MASs are set as $u_1(1)=u_2(1)=u_3(1)=u_4(1)=0$, $y_1(1)=-1$, $y_1(2)=1$, $y_2(1)=-1.35$, $y_2(2)=1.2$, $y_3(1)=-1.5$, $y_3(2)=1.3$, $y_4(1)=-1$, $y_4(2)=1.2$. Select the initial pseudo-partial-derivatives and step-size factors as $\hat{\psi}_1(1)=2$, $\hat{\psi}_2(1)=2.2$, $\hat{\psi}_3(1)=2.3$, $\hat{\psi}_4(1)=1.9$, $\rho_1(1)=0.25$, $\rho_2(1)=0.3$, $\rho_3(1)=0.1$ and $\rho_4(1)=0.15$, respectively. The remaining controller parameters are selected as $\mu=0.5$, $\lambda=0.3$, and $\sigma=10^{-5}$. It is worth pointing out that the Laplace matrix information has not been used in the design of controller parameters.

The proposed control algorithms are implemented to the MASs with four different communication graphs depicted in Fig. 2, and the corresponding system output response curves are plotted in Figs. 3-6 (a). Obviously, the designed distributed MFAC algorithms can achieve the consensus among agents. The corresponding triggering instants are recorded in Figs. 3-6 (b), from which we can see that before 80s, the triggering frequency promptly increases when the system becomes unstable. After 80s, the internal execution time is appropriately extended to save more communication resources when the system becomes stable. The adaptive step-size operators and auxiliary factors are shown in Fig. 7, which can automatically converge to the appropriate values, making it different from the existing results [25–32], where the step-size operator is a constant and needs to be artificially selected according to the system characteristics and communication topology structure.



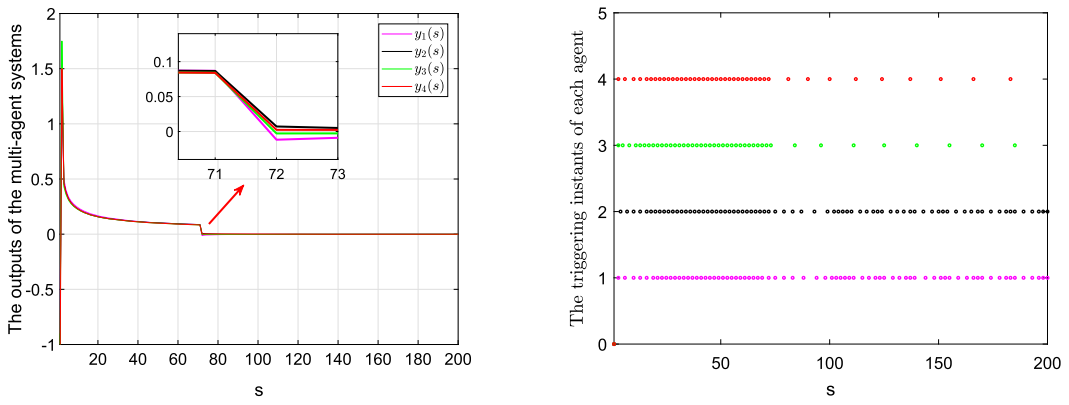
(a) The output responses (b) The event-triggered instants

Fig. 4. The output responses of the MASs with the communication topology I.



(a) The output responses (b) The event-triggered instants

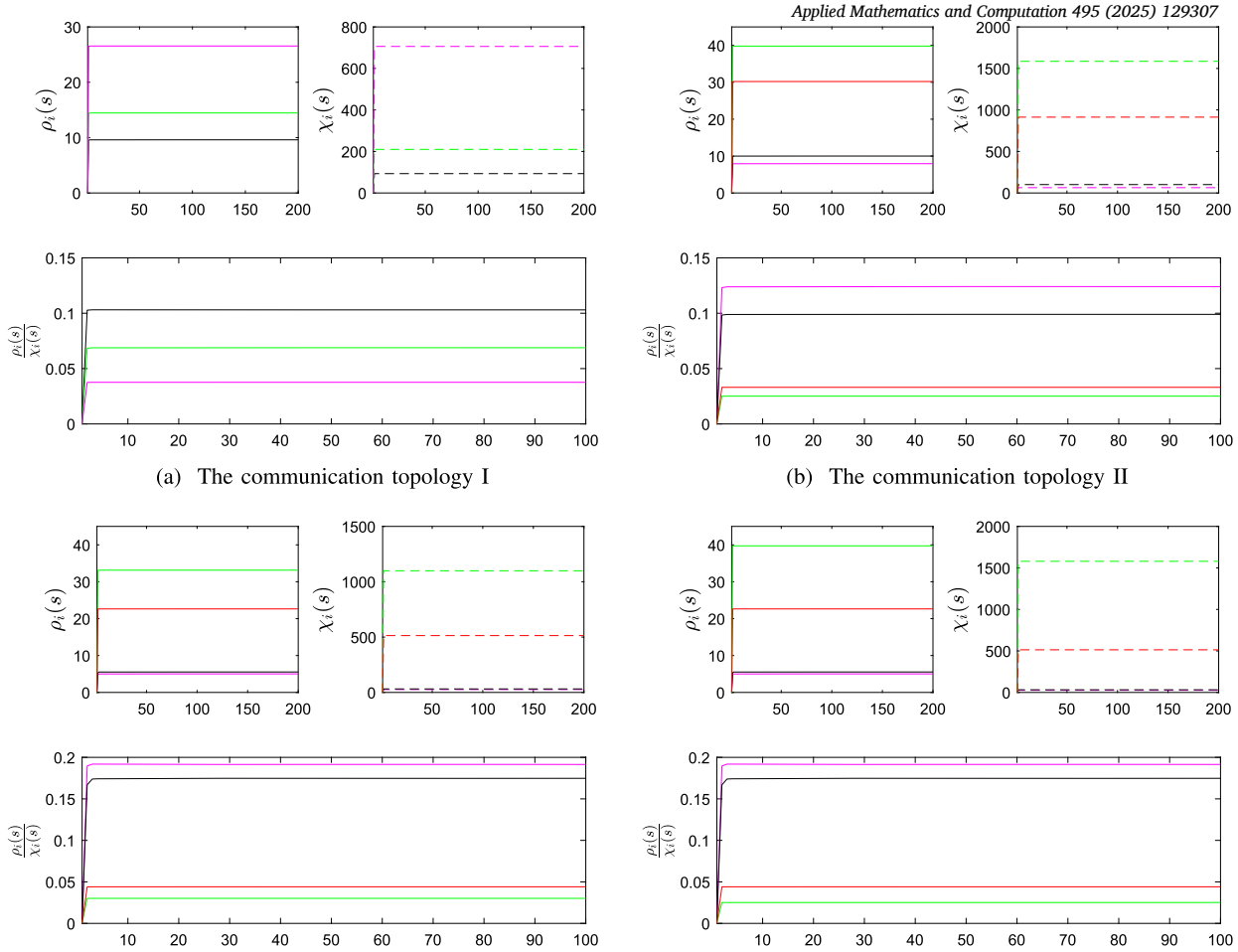
Fig. 5. The output responses of the MASs with the communication topology III.



(a) The output responses (b) The event-triggered instants

Fig. 6. The output responses of the MASs with the communication topology IV.

For the control strategy without ETM, data will be transmitted at each sampling instant. To illustrate the effectiveness of ETM (3), its transmission percentages are presented in Table 1, where the adopted ETM significantly reduces the communication burden. To demonstrate the superiority of the transmission mechanism (3), Table 1 shows the comparison of the static ETM [33–35] on event-triggered frequency. It can be observed that the dynamic ETM proposed in this paper is more effective in saving the communication resource.



(a) The communication topology I

(b) The communication topology II

(c) The communication topology III

(d) The communication topology IV

Fig. 7. The curves of parameters $\rho_i(s)$ and $\chi_i(s)$.

6. Conclusion

In this article, a learning-based fully distributed MFAC framework has been constructed for nonlinear MASs by using the dynamical linearization approach. A novel control algorithm with adaptive step-size factors was proposed to achieve consensus among agents. Furthermore, a dynamic ETM with a switch-adjustable threshold parameter has been incorporated into the framework to conserve network resources more efficiently. It has been proven that the consensus errors of systems converge asymptotically to zero under the distributed event-triggered MFAC algorithms. Finally, several simulation examples were provided to demonstrate the effectiveness of the proposed algorithms.

Data availability

No data was used for the research described in the article.

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